CHAPTER 03

MOVEMENT OF SUBSTANCES INTO OR OUT OF THE CELLS

3 Movement into and out of cells

3.1 Diffusion

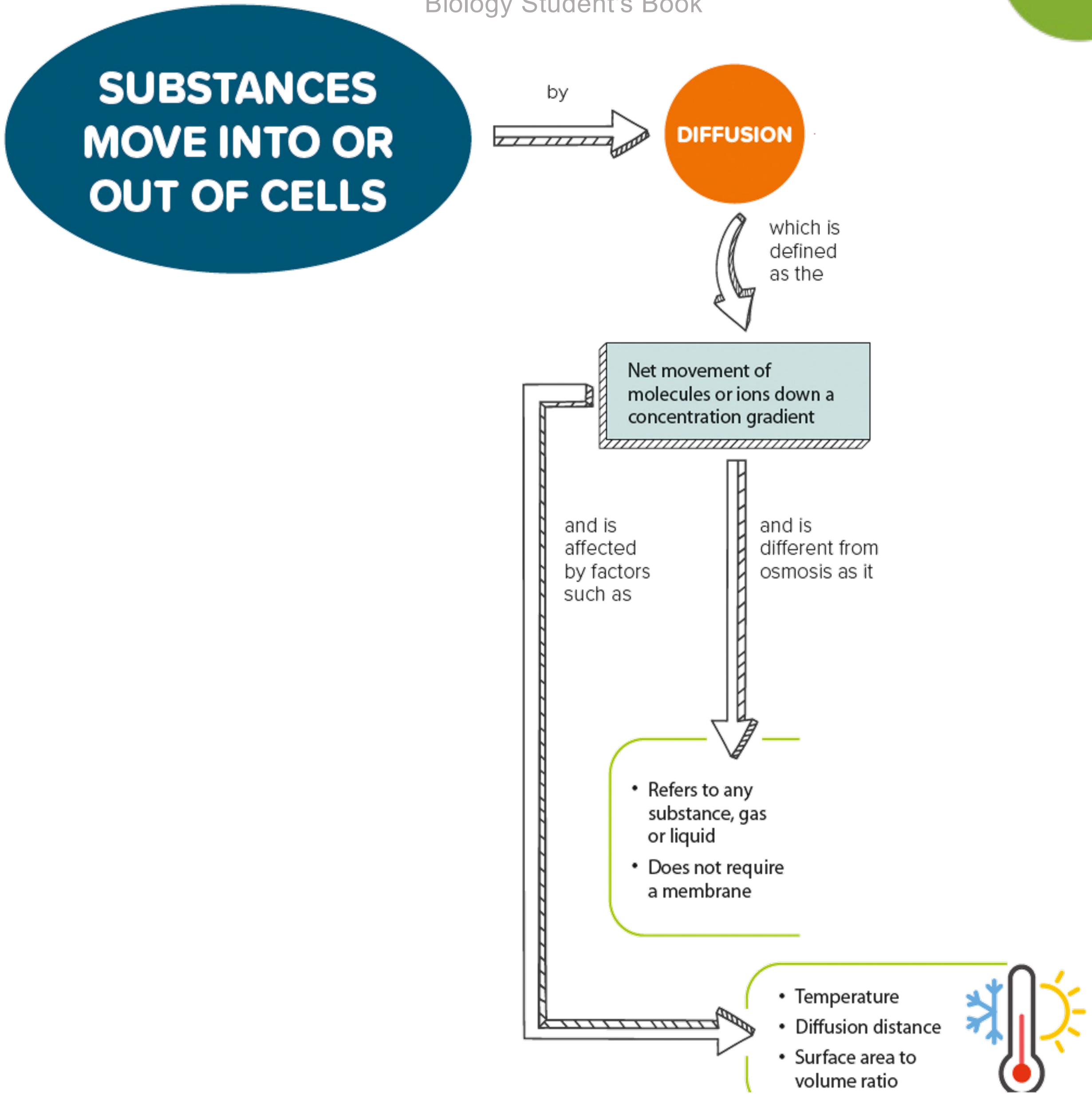
Core

1. Describe diffusion as the net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement
2. State that the energy for diffusion comes from the kinetic energy of random movement of molecules and ions
3. State that some substances move into and out of cells by diffusion through the cell membrane
4. Describe the importance of diffusion of gases and solutes in living organisms
5. Investigate the factors that influence diffusion, limited to: surface area, temperature, concentration gradient and distance

3.1 Diffusion

In this section, you will learn the following:

- Describe diffusion.
- State that the energy for diffusion comes from kinetic energy.
- State that some substances move into and out of cells by diffusion through the cell membrane.
- Describe the importance of diffusion of gases and solutes in living organisms.
- Investigate some factors that influence diffusion.



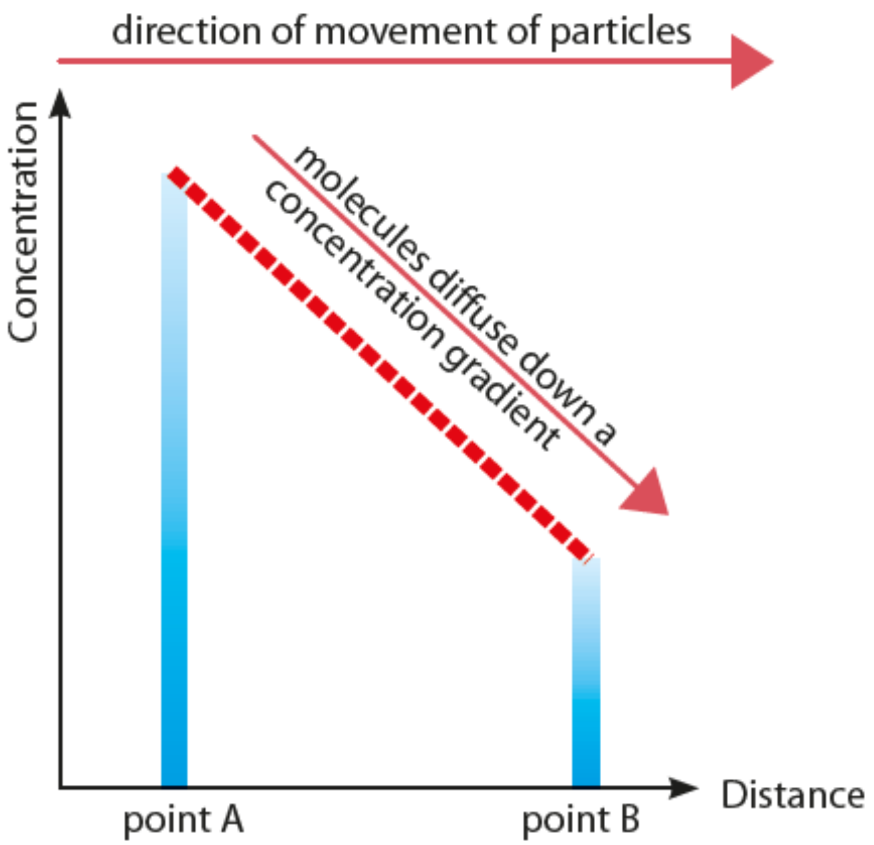
What is diffusion?

Diffusion of perfume particles
(Figure 3.1 of Student's Book)

1. All matter, including perfume, is made up of particles such as atoms, molecules or ions.
2. Individual particles of perfume evaporate to become a gas.
3. The particles of the gas are constantly moving. **Kinetic energy** causes the particles to move.
4. The gas spreads outwards through a process called **diffusion**.

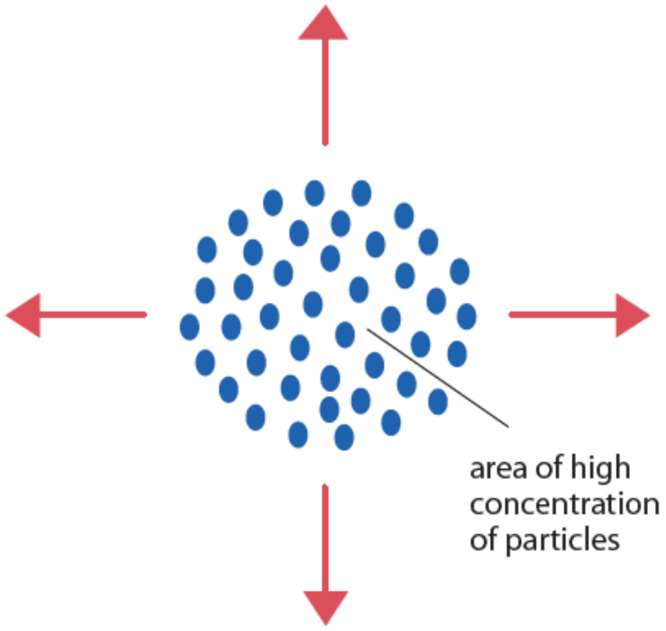
What is a concentration gradient and how is it related to diffusion?

- There are more particles at point A than at point B. Thus, there is a concentration difference between these two points.
- **Concentration gradient** is the difference in concentration between two regions.



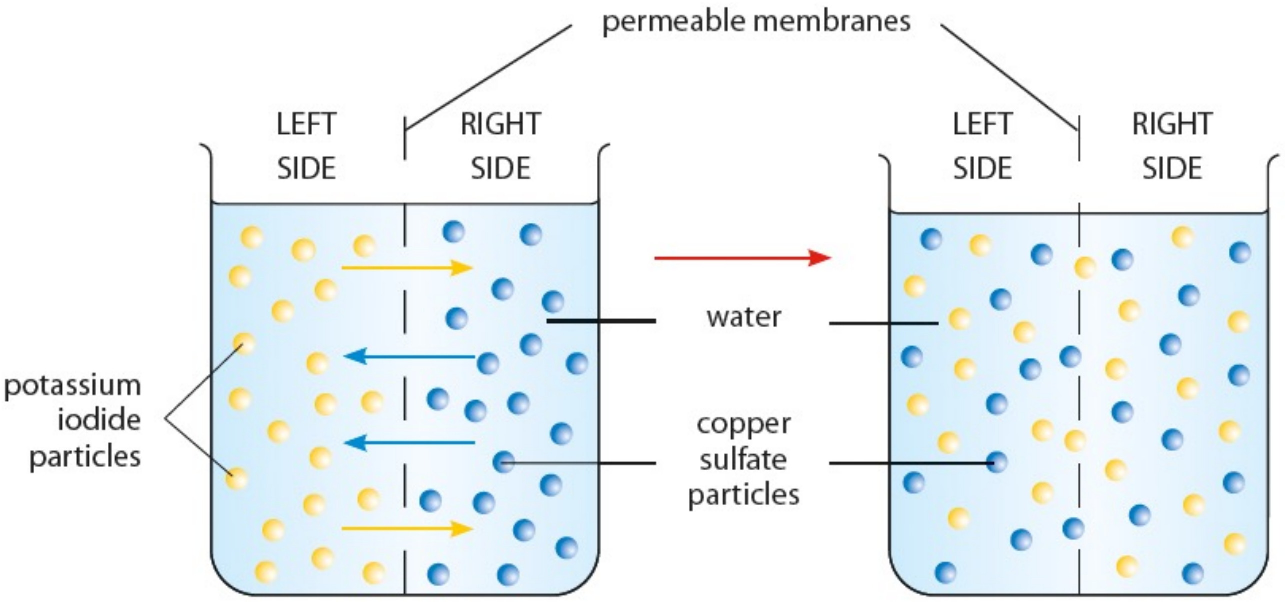
What is a concentration gradient and how is it related to diffusion?

- Randomly moving particles move (diffuse) down the concentration gradient and become evenly spaced out after some time.
- **Diffusion** is the net movement of particles from a region of their higher concentration to a region of their lower concentration, down a concentration gradient, as a result of random movement.



Arrows show direction of movement of particles (Figure 3.3 of Student's Book)

Can particles diffuse across a membrane?

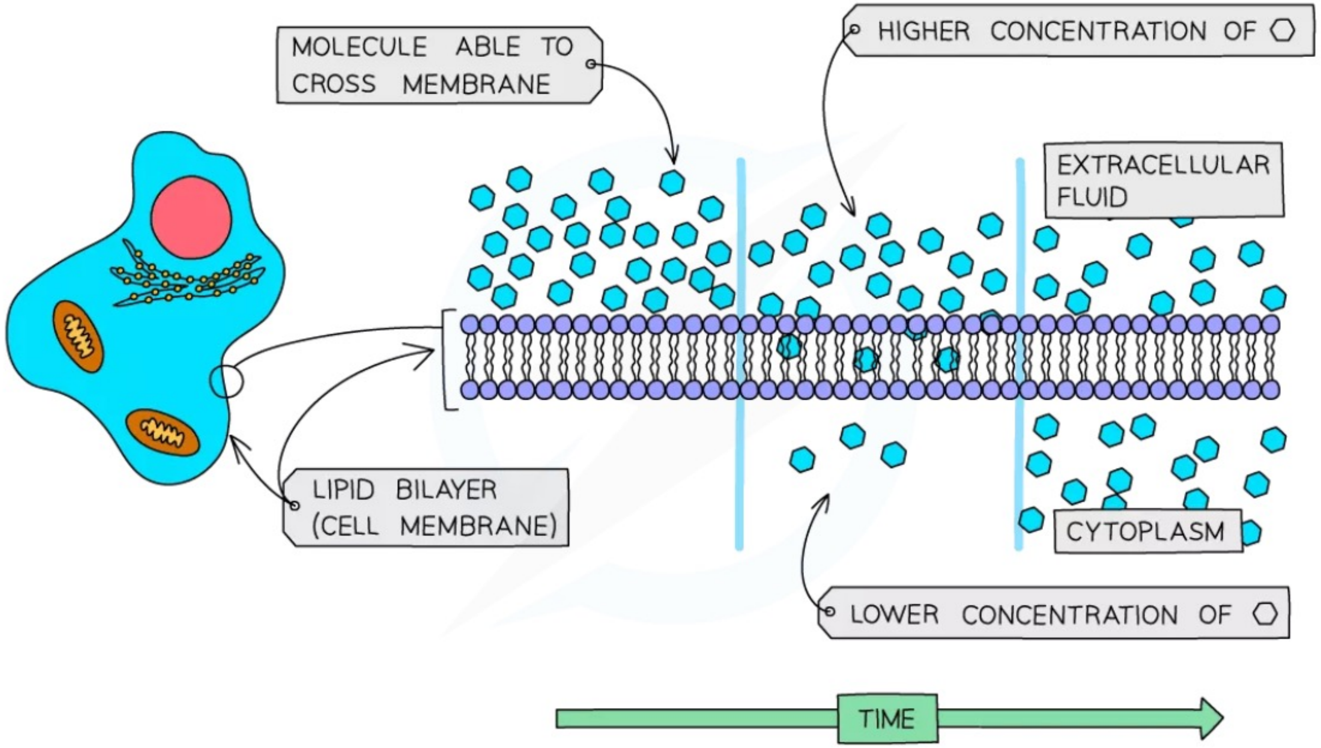


Diffusion of potassium iodide and copper sulfate particles through a permeable membrane (Figure 3.5 of Student's Book)

- Diffusion can occur across a permeable membrane.
- Eventually, there will be equal concentrations of all particles on both sides of the membrane.

Diffusion in Living Organisms

- Diffusion is the **movement of molecules** from a region of its **higher** concentration to a region of its **lower** concentration
- Molecules **move down a concentration gradient**, as a result of their **random movement**



- For **living cells**, the principle of the movement down a concentration gradient is the same, but the cell is surrounded by a **cell membrane** which can **restrict the free movement of the molecules**
- The cell membrane is a **partially permeable membrane** – this means it allows some molecules to cross easily, but others with difficulty or not at all
- The simplest sort of selection is based on the size of the molecules
- Diffusion helps living organisms to:
    - obtain many of their **requirements**
    - get rid of many of their **waste products**
    - carry out **gas exchange for respiration**

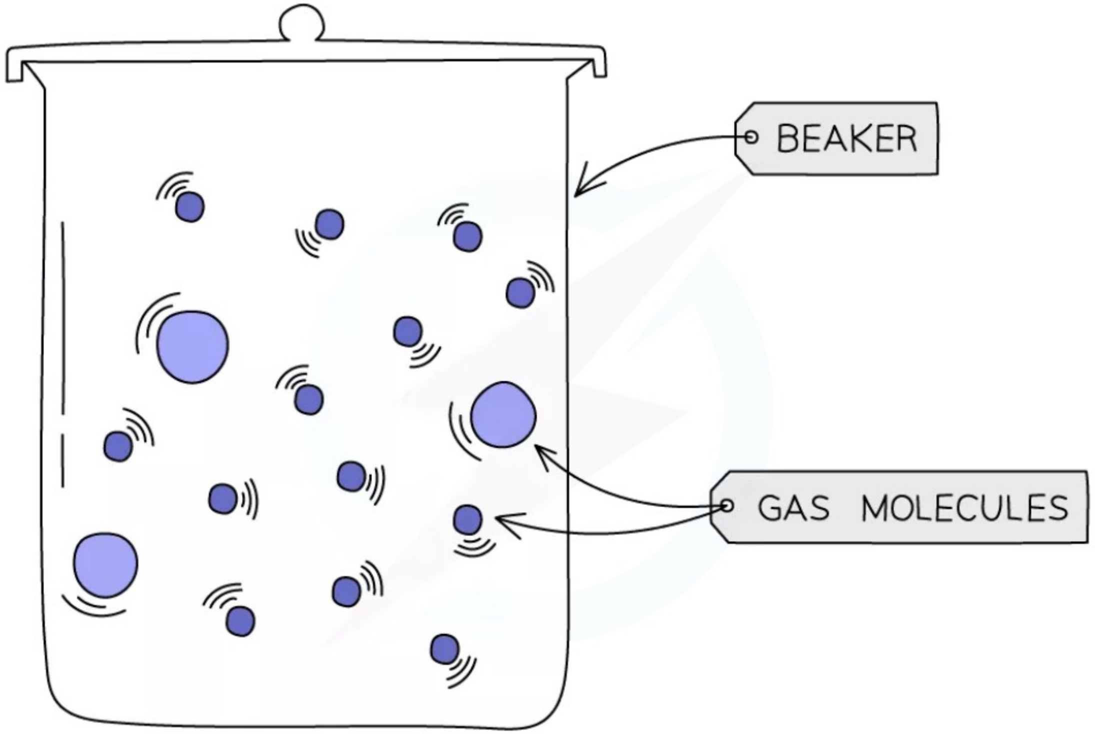
Examples of diffusion in living organisms

- You will need to learn examples of substances that organisms obtain by diffusion
- Don't forget that **plants require oxygen for respiration at all times**, as well as **carbon dioxide for photosynthesis** when conditions for photosynthesis are right (e.g. enough light and a suitable temperature)

SITE	MOLECULES MOVING	FROM	TO
SMALL INTESTINE	DIGESTED FOOD PRODUCTS – GLUCOSE, AMINO ACIDS, FATTY ACIDS AND GLYCEROL ETC.	LUMEN OF SMALL INTESTINE	BLOOD / LYMPH IN VILLI FOUND COVERING SMALL INTESTINE WALLS
LEAF	OXYGEN	AIR SPACES BETWEEN MESOPHYLL CELLS	MITOCHONDRIA IN ALL CELLS
LEAF	CARBON DIOXIDE	AIR SPACES BETWEEN MESOPHYLL CELLS	CHLOROPLASTS IN MESOPHYLL CELLS
LEAF	WATER VAPOUR	STOMATAL PORES	AIR OUTSIDE STOMATA
LUNGS	OXYGEN	ALVEOLAR AIR SPACE	BLOOD IN CAPILLARIES AROUND ALVEOLI
LUNGS	CARBON DIOXIDE	BLOOD IN CAPILLARIES AROUND ALVEOLI	ALVEOLAR AIR SPACE

Brownian Motion

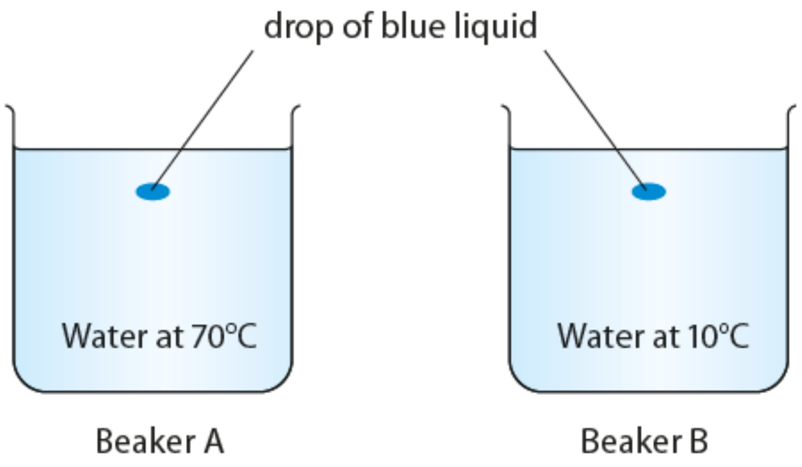
- All particles move randomly at all times
- This is known as Brownian motion
- The energy for diffusion comes from the kinetic energy of this random movement of molecules and ions



What are the factors that affect the rate of diffusion?

1. **Temperature**
   - The higher the temperature, the faster the rate of diffusion.

Let's Investigate 3B



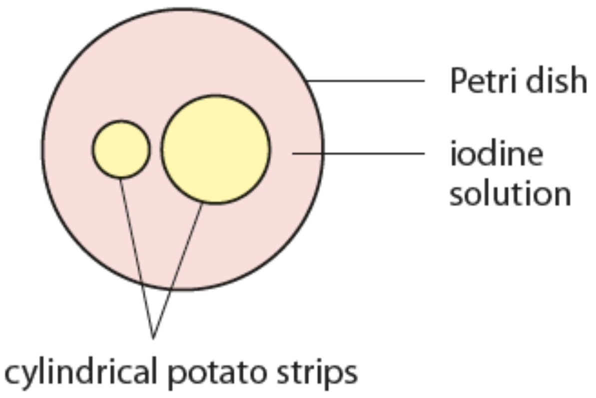
- The water in beaker A has a higher temperature than the water in beaker B.
- The blue particles in beaker A have more kinetic energy and thus diffuse at a faster rate.

What are the factors that affect the rate of diffusion?

2. Diffusion distance

- The shorter the diffusion distance, the faster the rate of diffusion.

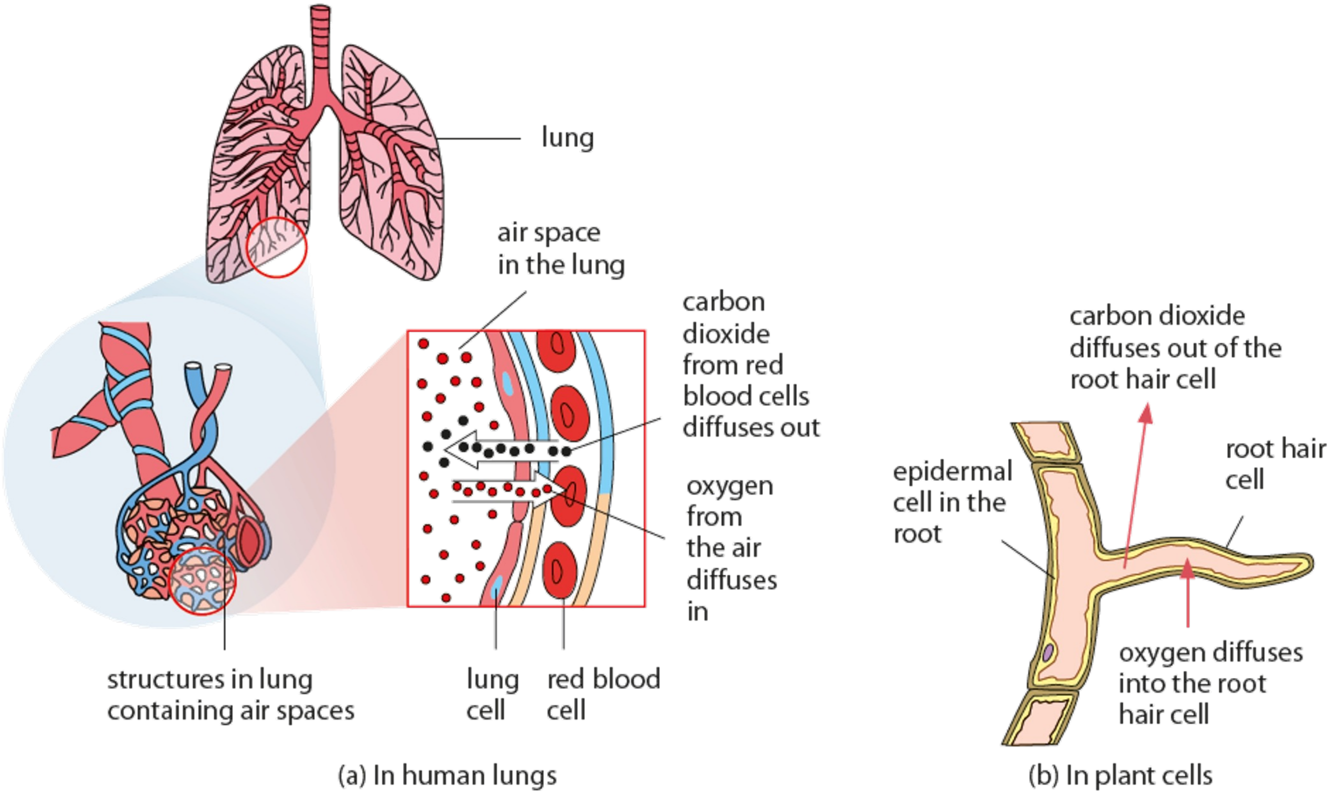
Let's Investigate 3C



- The surface of each potato strip will turn blue-black when it comes in contact with iodine solution.
- It takes more time for the blue-black colour to reach the centre of the thicker potato strip.
- The rate of diffusion is slower across longer diffusion distances.

What are the factors that affect the rate of diffusion?

- Diffusion is an important way by which oxygen and carbon dioxide move into and out of cells.
- Exchange surfaces have a short diffusion distance.



LINK

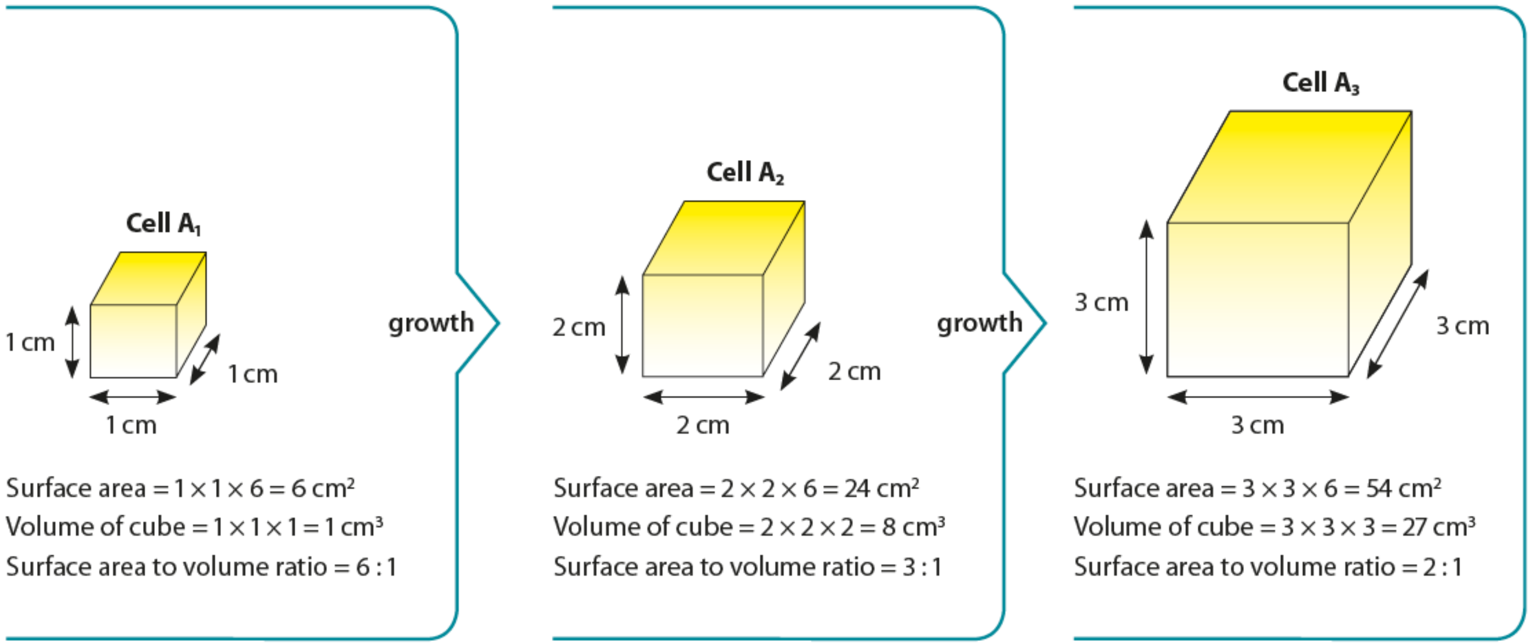
How do humans carry out gas exchange? Find out in Chapter 11.

How do cells of the small intestine take up nutrients? Find out in Chapter 7.

What are the factors that affect the rate of diffusion?

3. Surface area to volume ratio

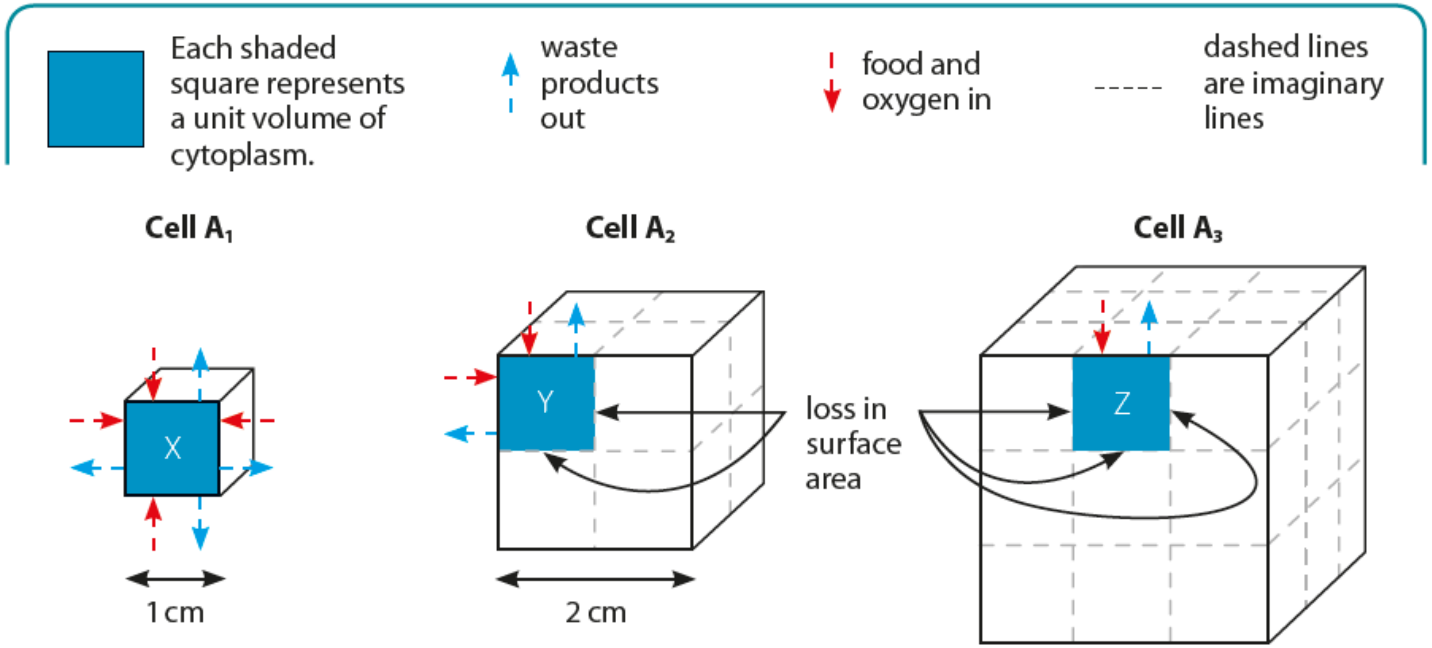
- The greater the area per unit volume, the faster the rate of diffusion of a substance for a given concentration gradient.
- Why do cells not grow beyond a certain size? Look at this example.



The increase in surface area is not as much as the increase in volume.

Surface area to volume ratio decreases.

What are the factors that affect the rate of diffusion?



Cell A_1 can obtain food and oxygen from all sides. It can carry out activities like growth efficiently.

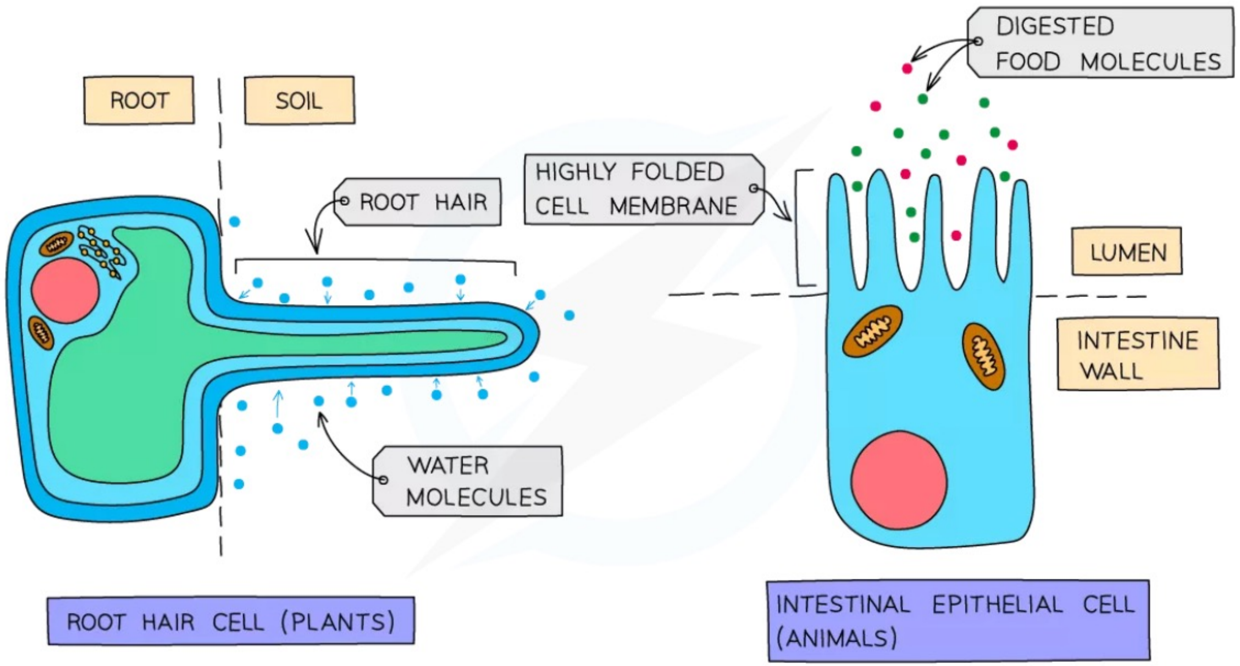
X and Y have similar volumes. Y has a reduced surface area, thus it receives nutrients and oxygen at a slower rate. Cell A_2 is less active than cell A_1.

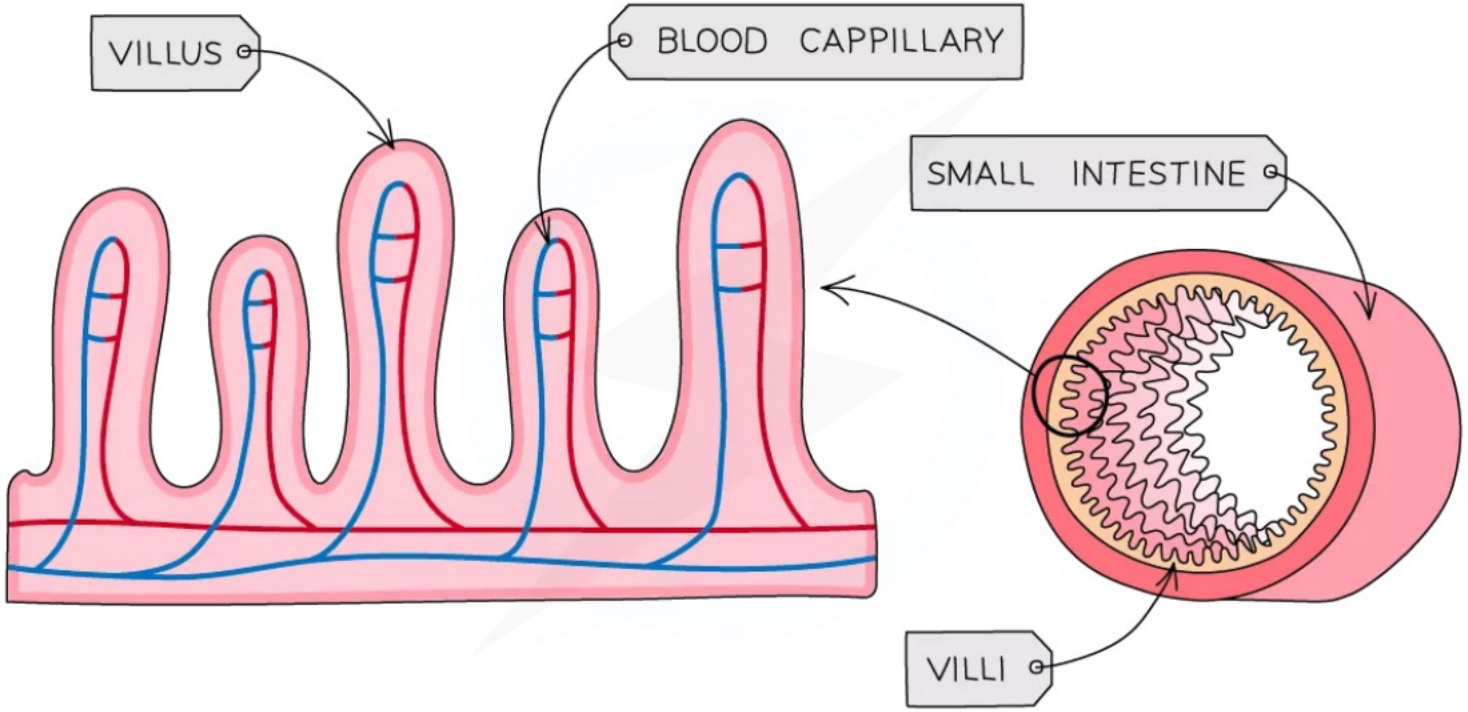
Z has the smallest surface area per unit volume. It gets less food and oxygen compared to X and Y. Cell A_3 is less active than A_1 and A_2.

Factors that Influence Diffusion

Surface area to volume ratio:

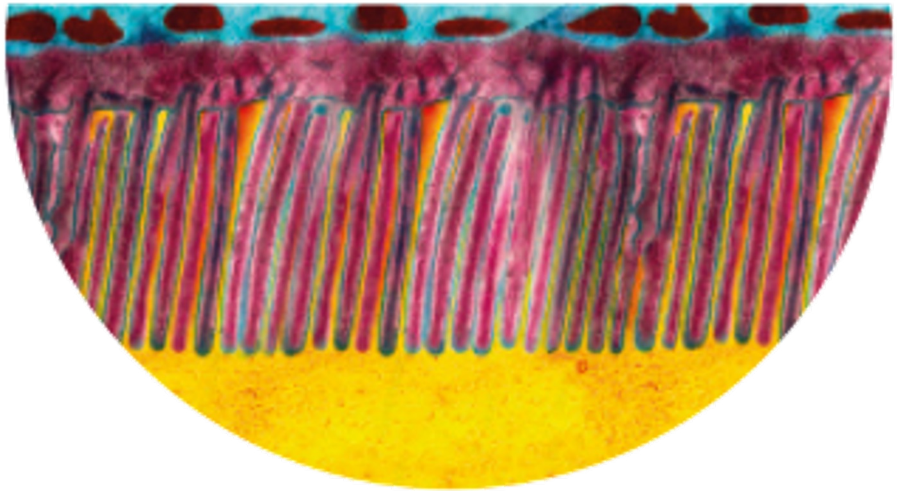
- The **bigger** a cell or structure is, the **smaller its surface area to volume ratio** is, slowing down the rate at which substances can move across its surface
- Many cells which are adapted for diffusion have **increased surface area** in some way – eg root hair cells in plants (which absorb water and mineral ions) and cells lining the ileum in animals (which absorb the products of digestion)




VILLUS
BLOOD CAPPILLARY
SMALL INTESTINE
VILLI

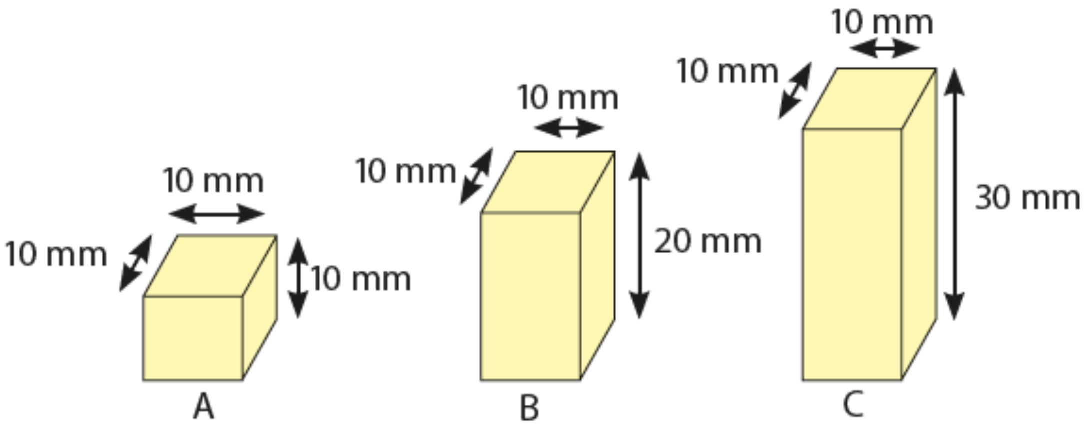
How do cells adapt their surface area for absorption?

- Some cells are adapted to absorb materials.
- These cells often have long, narrow protrusions or folds in their membranes.



Colourised electron micrograph showing highly magnified microvilli on the edge of an intestine epithelial cell

Let's Investigate 3D



- Agar block A has the largest surface area to volume ratio. It takes the shortest time to turn pink. Thus, the rate of absorption increases as surface area to volume ratio increases.
- Cells in the body should have the shape of agar block A to increase the rate of substances entering or leaving the cells.

Three factors that affect the rate of diffusion and therefore the movement of molecules through membranes:

Distance

- The smaller the distance molecules have to travel the faster transport will occur
- This is why blood capillaries and alveoli have walls which are only one cell thick, ensure the rate of diffusion across them is as fast as possible

Temperature

- The higher the temperature, the faster molecules move as they have more energy
- This results in more collisions against the cell membrane and therefore a faster rate of movement across them

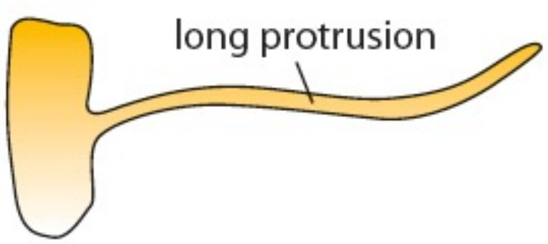
Concentration Gradient

- The greater the difference in concentration either side of the membrane, the faster movement across it will occur
- This is because on the side with the higher concentration, more random collisions against the membrane will occur

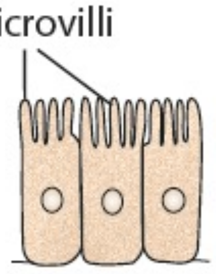
Let's Practise 3.1

a) How are the root hair cell, the epithelial cell of the small intestine and the red blood cell adapted for absorption of materials?

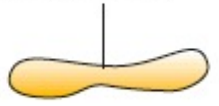
b) What feature(s) do the three cells have in common?



Root hair cell



Epithelial cell



Red blood cell

3.2 Osmosis

Core

1. Describe the role of water as a solvent in organisms with reference to digestion, excretion and transport
2. State that water diffuses through partially permeable membranes by osmosis
3. State that water moves into and out of cells by osmosis through the cell membrane
4. Investigate osmosis using materials such as dialysis tubing
5. Investigate and describe the effects on plant tissues of immersing them in solutions of different concentrations
6. State that plants are supported by the pressure of water inside the cells pressing outwards on the cell wall

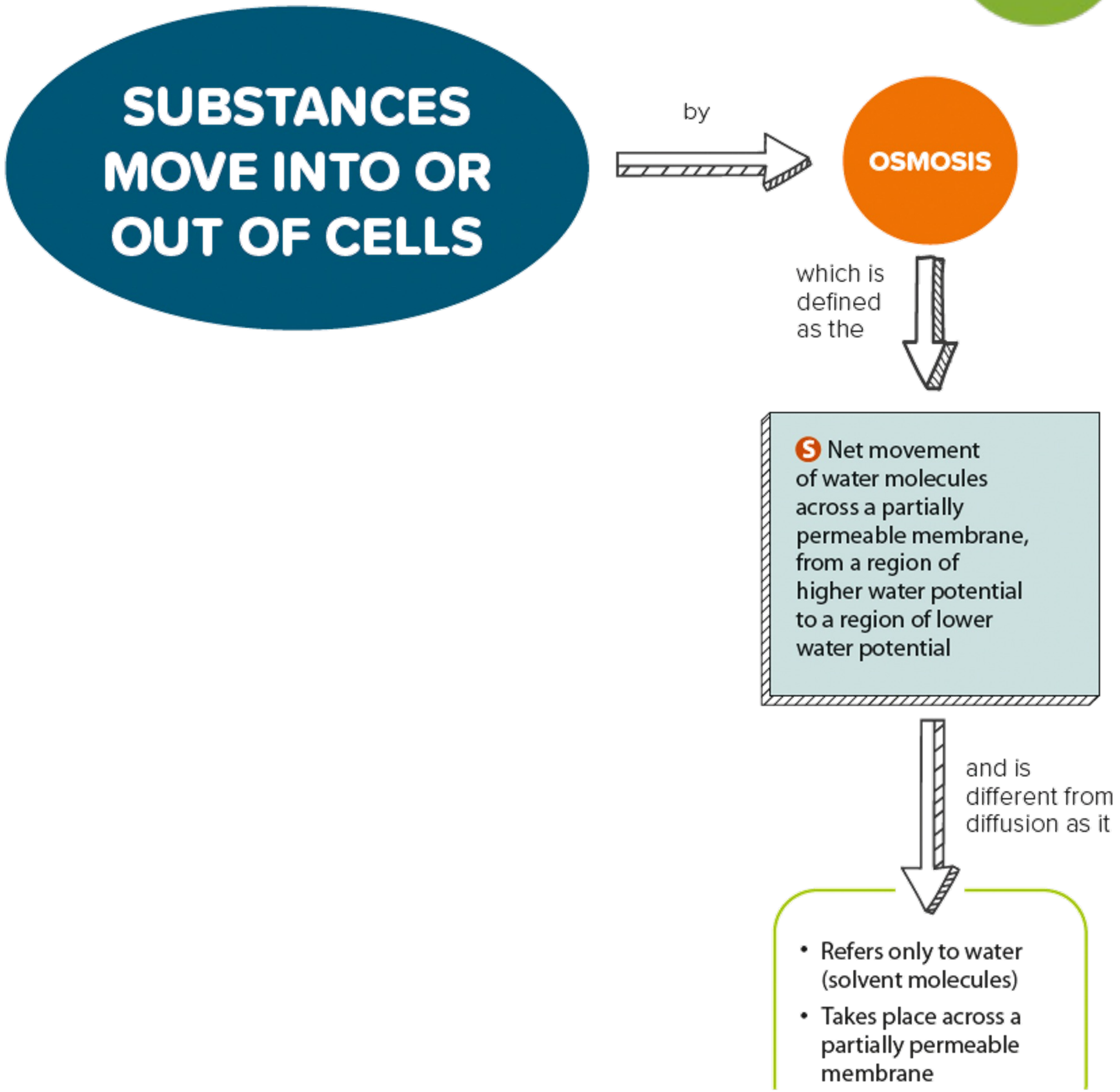
Supplement

7. Describe osmosis as the net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution), through a partially permeable membrane
8. Explain the effects on plant cells of immersing them in solutions of different concentrations by using the terms: turgid, turgor pressure, plasmolysis, flaccid
9. Explain the importance of water potential and osmosis in the uptake and loss of water by organisms

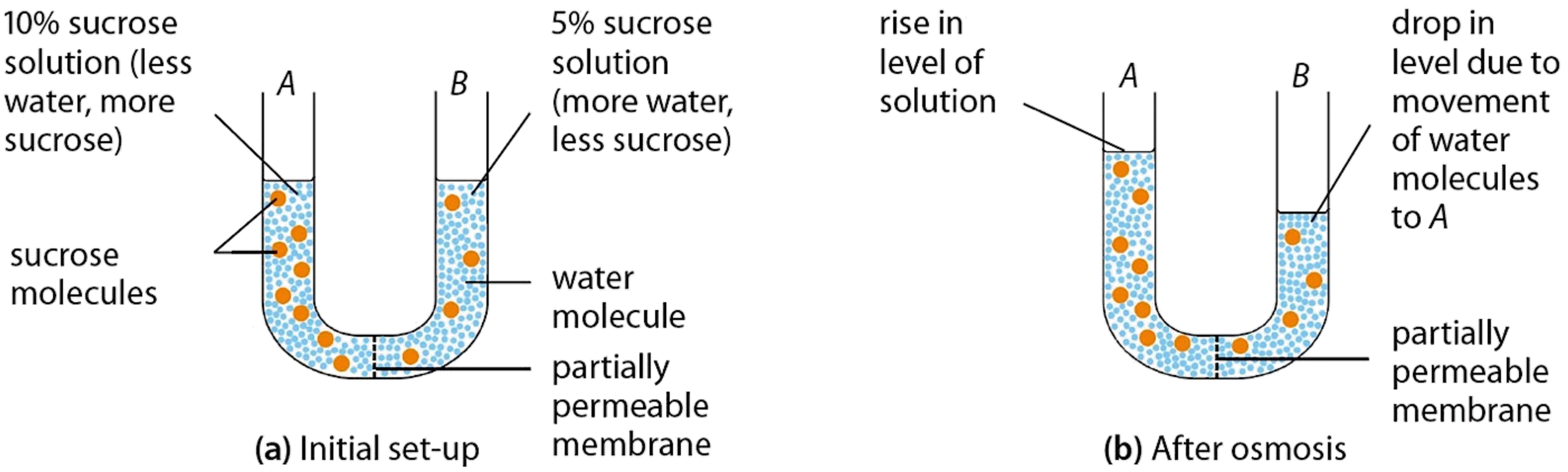
3.2 Osmosis

In this section, you will learn the following:

- Describe the role of water as a solvent in digestion, excretion and transport.
- S Describe osmosis.
- State that water diffuses through partially permeable membranes by osmosis.
- State that water moves into and out of cells by osmosis through the cell membrane.
- Investigate osmosis using dialysis tubing.
- Investigate and describe the effect of immersing plant tissues in solutions of different concentrations.
- S Explain the effects of immersing plant cells in solutions of different concentrations.
- S Explain the importance of water potential and osmosis in the uptake and loss of water by organisms.
- State that plants are supported by water pressure inside the cells pressing outwards on the cell wall.



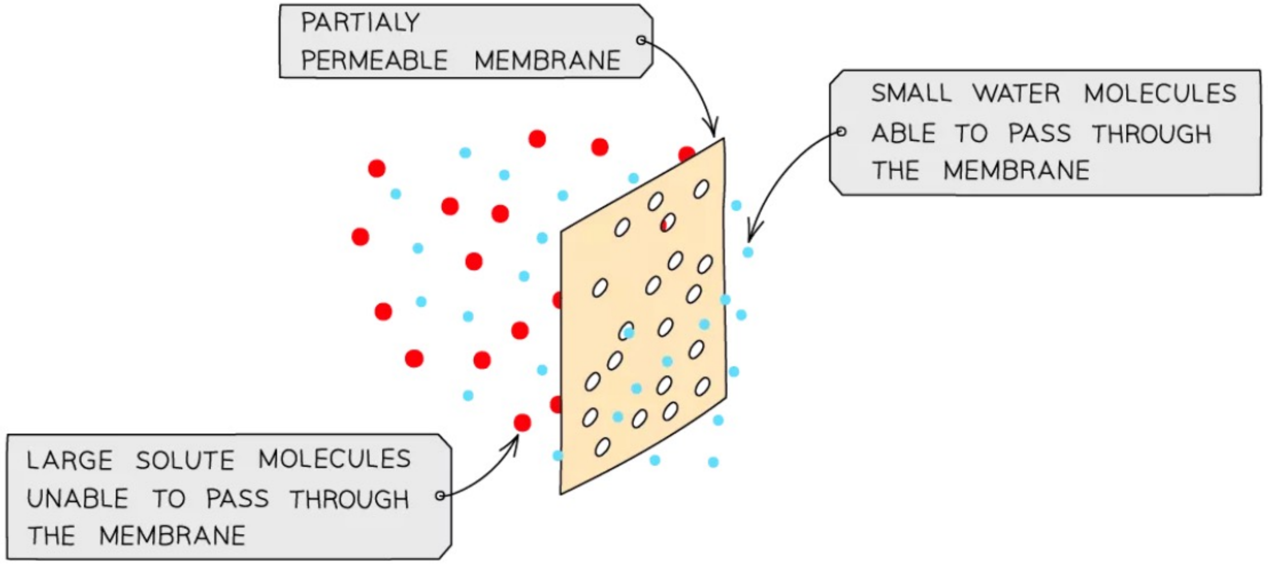
What is osmosis?



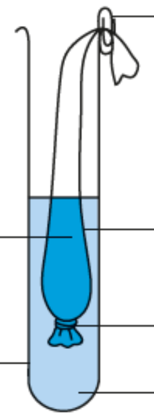
- The membrane is permeable to water molecules but not to sucrose molecules.
- Water molecules diffuse from B to A because B has a higher concentration of water than A.
- S Water molecules diffuse from B to A because A has a lower water potential than B.
- **Osmosis** is the diffusion of water molecules across a partially permeable membrane. Water moves into and out of cells by osmosis through the cell membrane.

Osmosis Theory: Basics

- All cells are surrounded by a cell membrane which is **partially permeable**
- **Water** can move in and out of cells by osmosis
- Osmosis is the diffusion of water molecules from a **dilute solution** (high concentration of water) to a **more concentrated solution** (low concentration of water) across a **partially permeable membrane**
- In doing this, water is moving down its **concentration gradient**
- The cell membrane is partially permeable which means it allows **small molecules** (like water) through but not larger molecules (like solute molecules)



Let's Investigate 3E

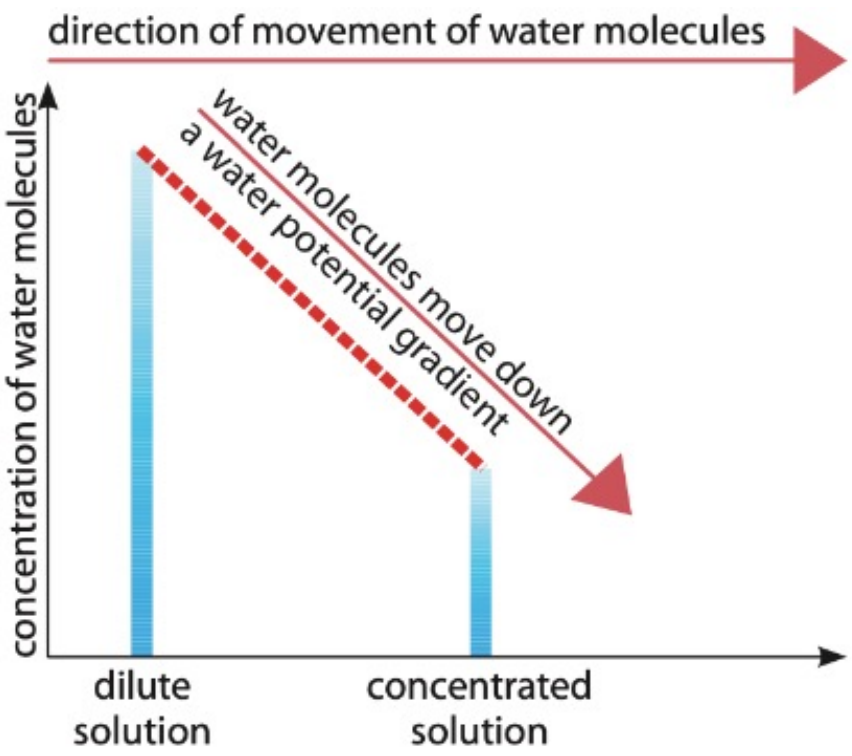


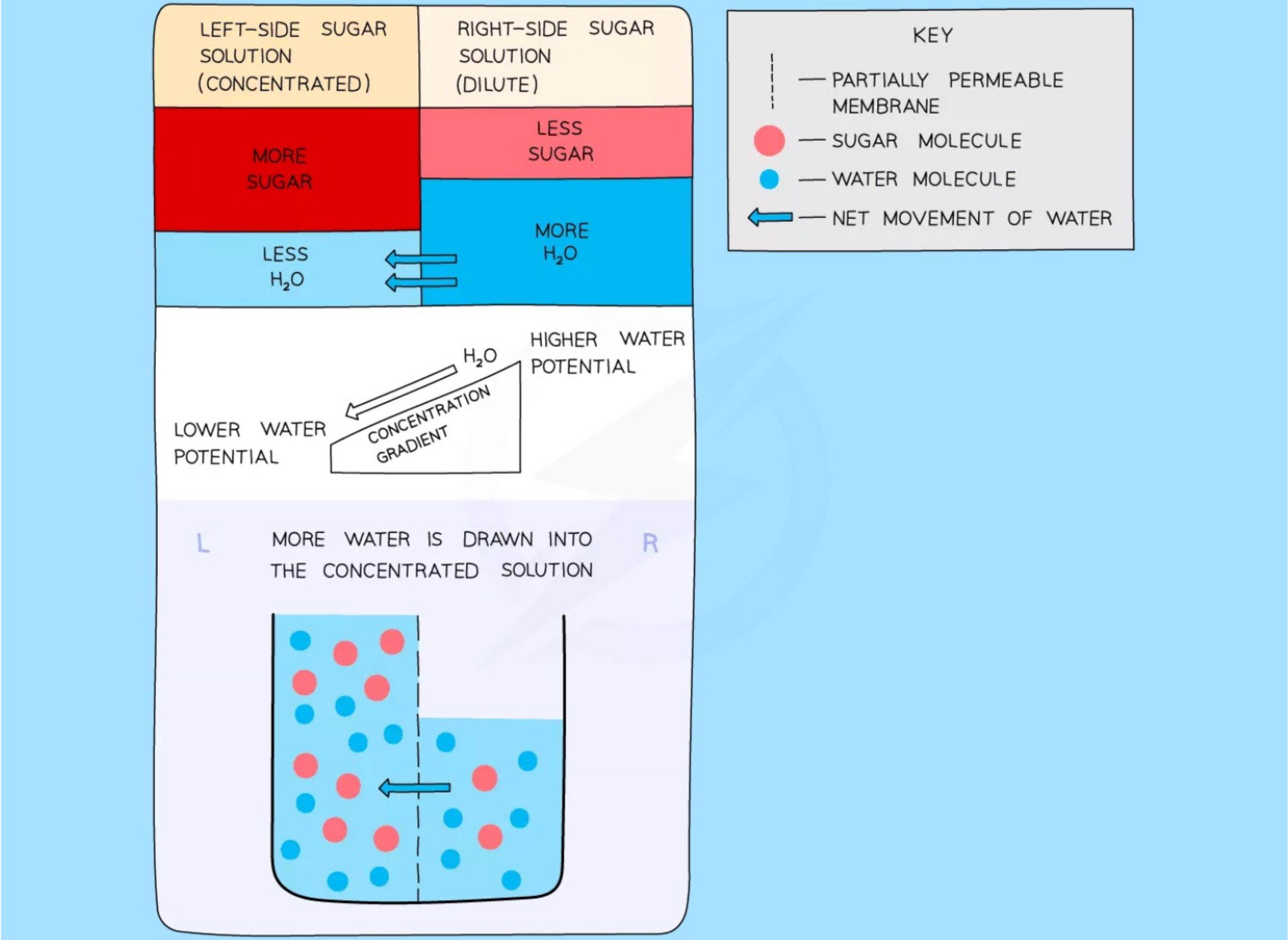
- The dialysis tubing is a partially permeable membrane.
- The concentration of water molecules is higher outside the tubing than in the tubing. Water enters the tubing by osmosis. So, the level of sucrose solution in the dialysis tubing rises.

S

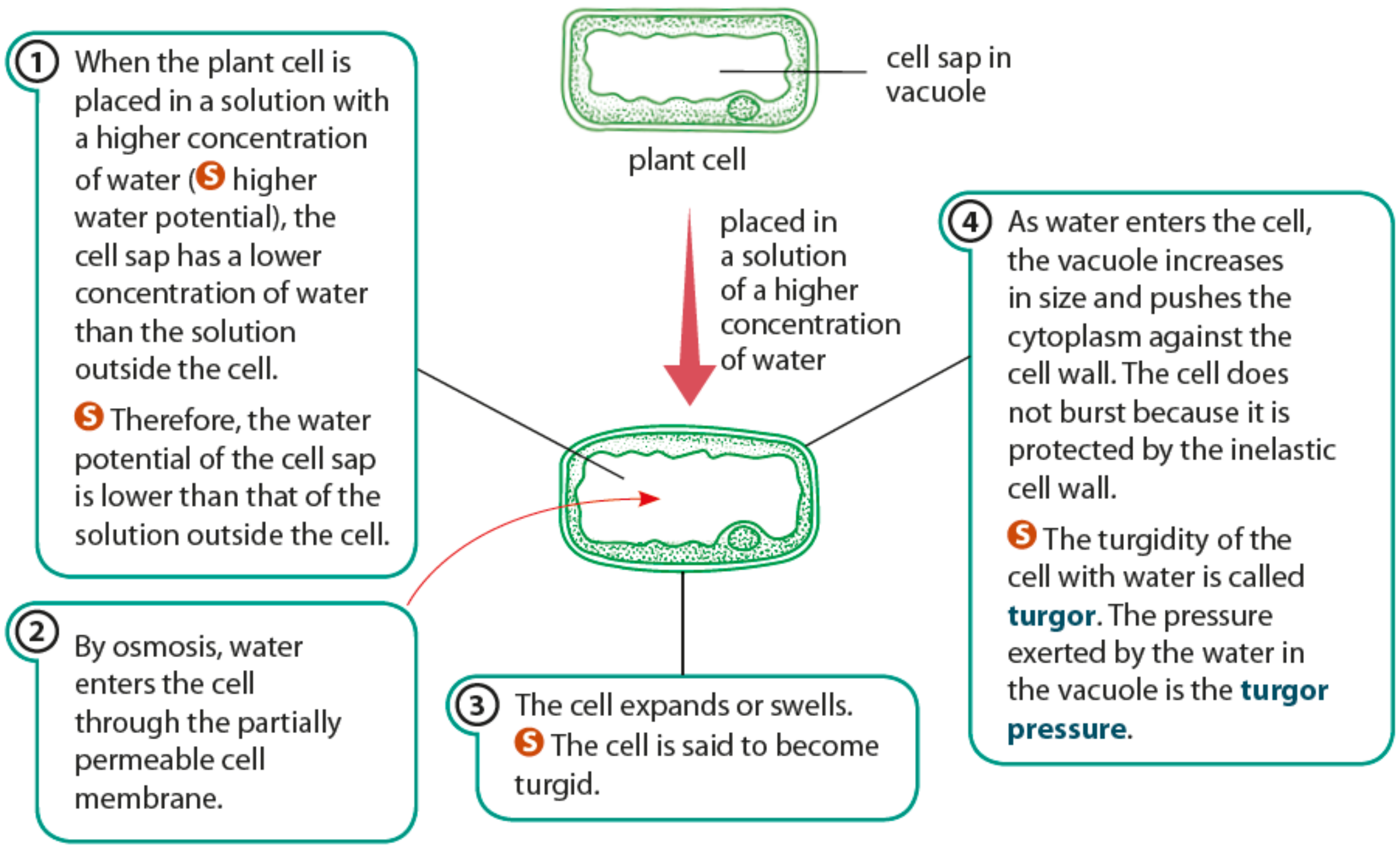
What is water potential and how is it related to osmosis?

- A dilute solution has a higher water potential than a concentrated solution for the same volume.
- **Osmosis** is the net movement of water molecules through a partially permeable membrane, from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution).




LEFT–SIDE SUGAR SOLUTION (CONCENTRATED)
RIGHT–SIDE SUGAR SOLUTION (DILUTE)
MORE SUGAR
LESS SUGAR
LESS H_2O
MORE H_2O
HIGHER WATER POTENTIAL
H_2O
CONCENTRATION GRADIENT
LOWER WATER POTENTIAL
L
MORE WATER IS DRAWN INTO THE CONCENTRATED SOLUTION
R
KEY
PARTIALLY PERMEABLE MEMBRANE
SUGAR MOLECULE
WATER MOLECULE
NET MOVEMENT OF WATER

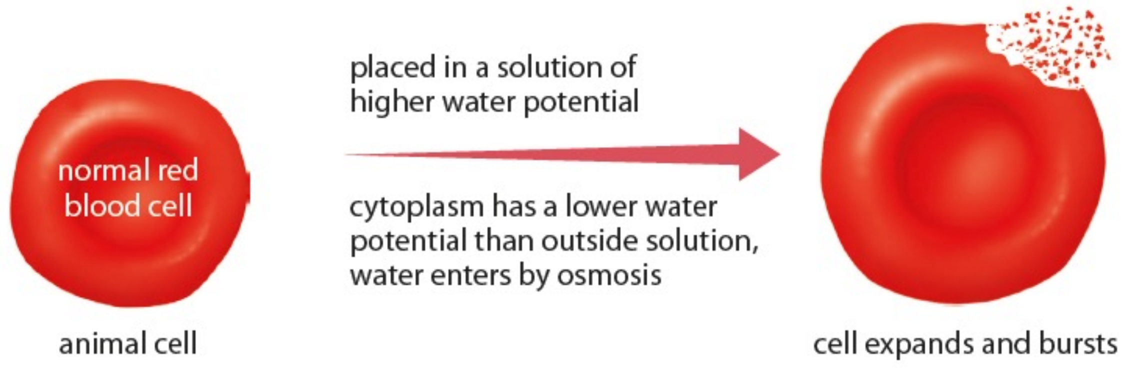
What happens to a plant cell in a solution with a higher concentration of water?



A plant cell in a solution with a higher concentration of water (Figure 3.24 of Student's Book)

What happens to an animal cell in a solution of higher water potential?

- An animal cell does not have a cell wall.
- The cell will swell and may even burst in a solution of higher water potential than the cytoplasm.

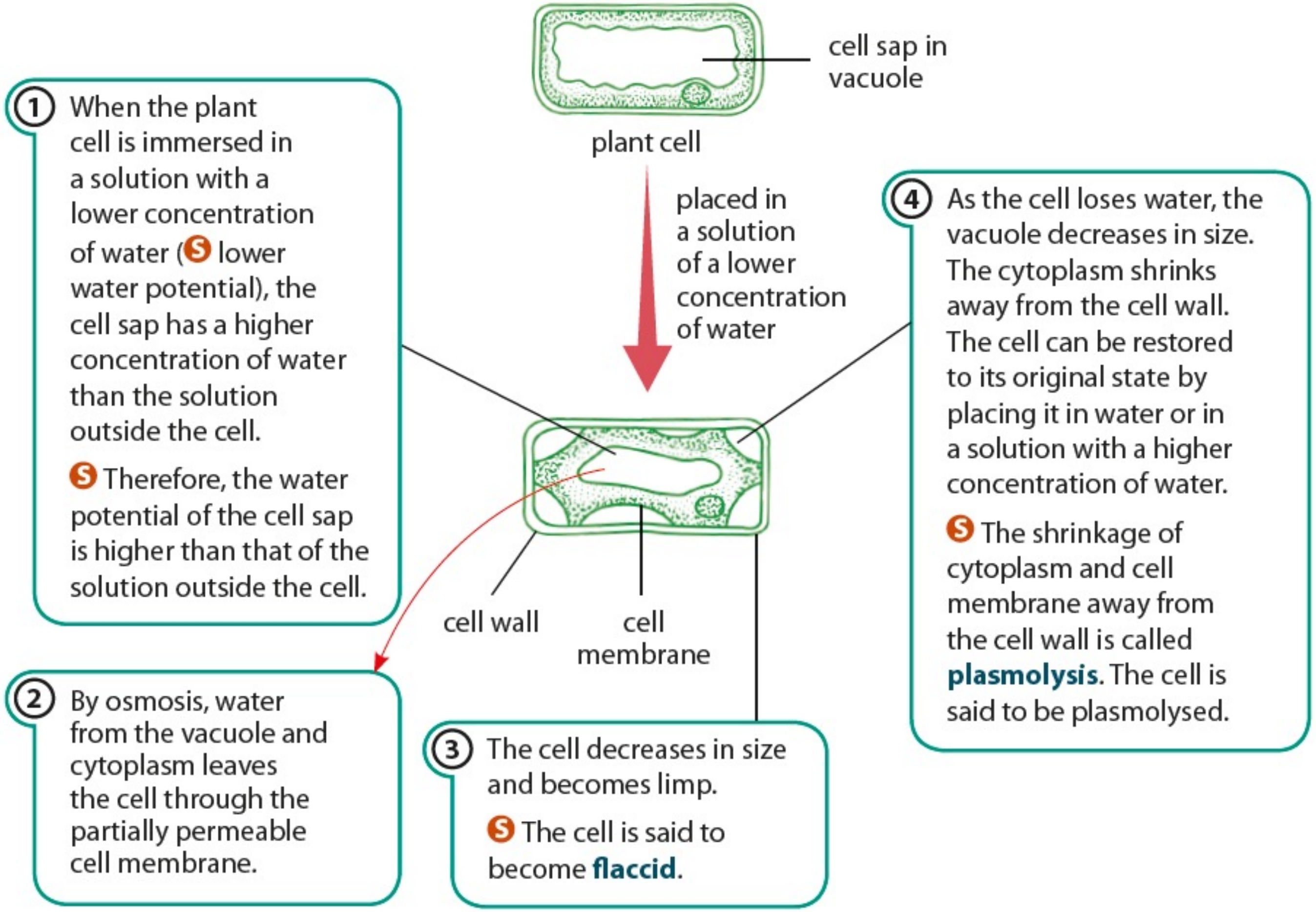


An animal cell in a solution with higher water potential

What happens to a cell in a solution of the same water potential?

- The cell will not change their size or shape.

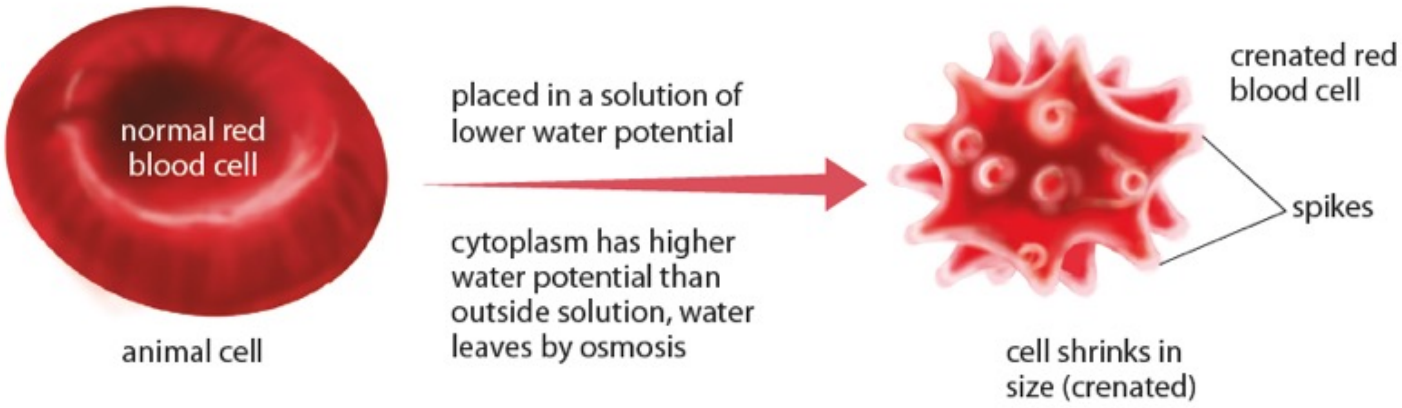
What happens to a plant cell in a solution with a lower concentration of water?



A plant cell in a solution with a lower concentration of water (Figure 3.24 of Student's Book)

S What happens to an animal cell in a solution of lower water potential?

- The cell shrinks and little spikes appear on the cell membrane. This process is called **crenation**.
- The cell will become dehydrated and eventually die.



An animal cell in a solution with lower water potential

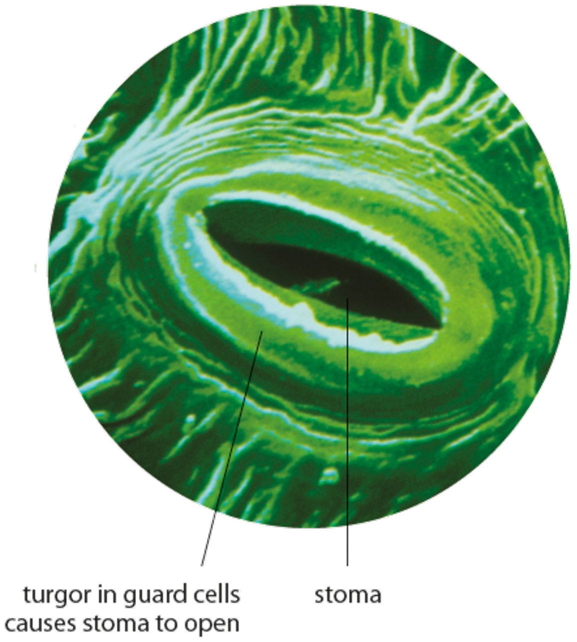
S

Why is turgor important in plants?

- Turgor is important for maintaining the shape of soft tissues in plants.
- Changes in the turgor of the guard cells cause the opening and closing of the stomata.
- Plasmolysis causes cells to become limp or flaccid.

Turgor pressure in the cells of the plant helps keep this plant firm and upright.

Loss of turgor due to excessive water loss may cause a plant to wilt.



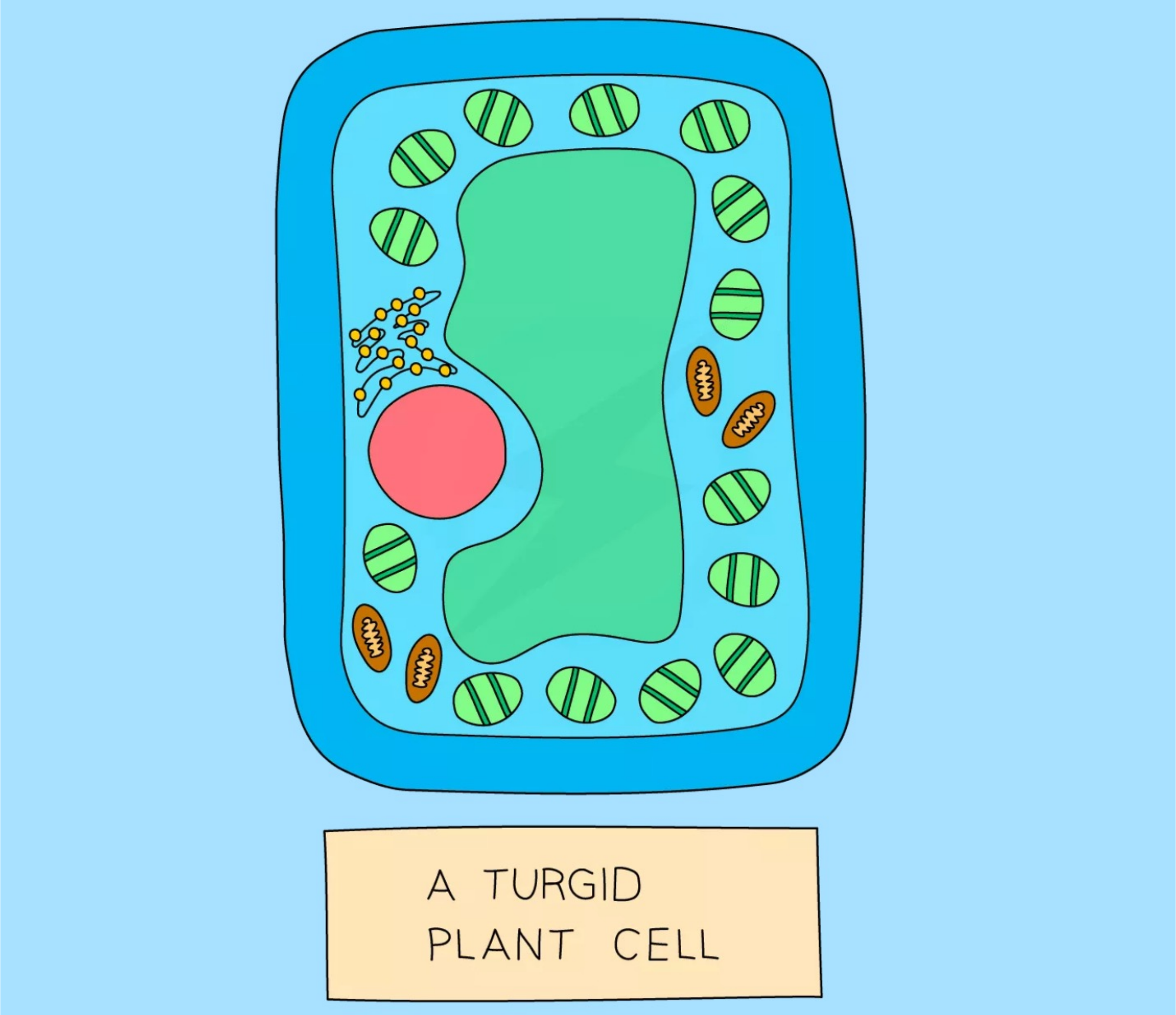
Electron micrograph of stomata

What are the differences between diffusion and osmosis?

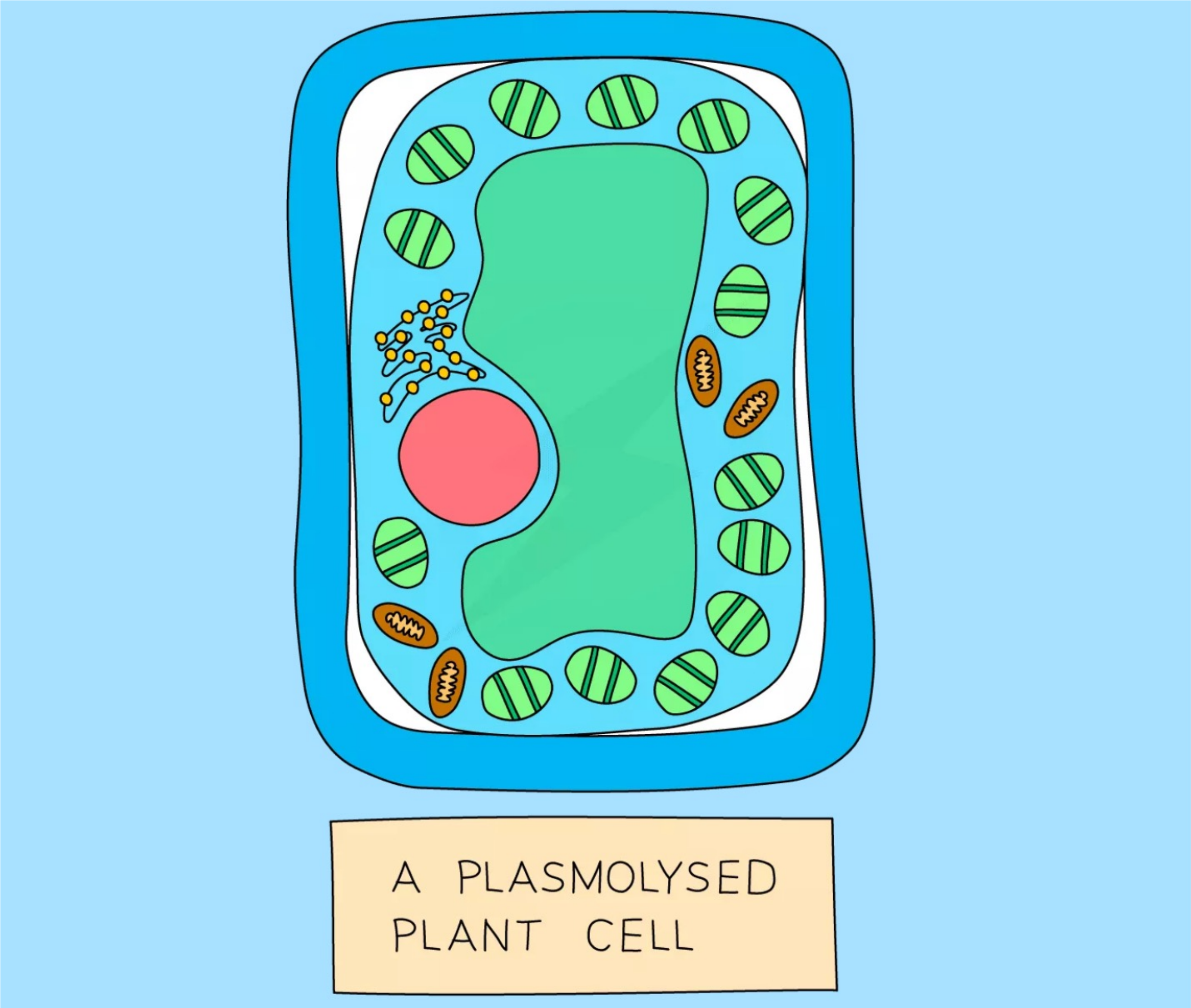
Diffusion	Osmosis
Net movement of a substance, gaseous or solute	Net movement of water molecules
Movement of particles down a concentration gradient	Movement of water molecules from a dilute solution to a more concentrated solution (S down a water potential gradient)
Partially permeable membrane not required	Partially permeable membrane required

Osmosis Experiments

- The most common osmosis practical involves cutting **cylinders of potato** and placing them into **distilled water** and **sucrose solutions** of increasing concentration
- The potato cylinders are weighed before placing into the solutions
- They are left in the solutions for 20 – 30 minutes and then removed, dried to remove excess liquid and reweighed
- The potato cylinder in the distilled water will have **increased its mass** the most as there is a greater concentration gradient in this tube between the distilled water (high water potential) and the potato cells (lower water potential)
- This means more water molecules will move **into the potato cells by osmosis**, pushing the cell membrane against the cell wall and so increasing the **turgor pressure** in the cells which makes them **turgid** – the potato cylinders will feel hard


A TURGID
PLANT CELL

- The potato cylinder in the strongest sucrose concentration will have **decreased its mass the most** as there is a greater concentration gradient in this tube between the potato cells (higher water potential) and the sucrose solution (lower water potential)
- This means more water molecules will move **out of the potato cells by osmosis**, making them flaccid and decreasing the mass of the cylinder – the potato cylinders will feel floppy
- If looked at underneath the microscope, cells from this potato cylinder might be **plasmolysed**, meaning the cell membrane has pulled away from the cell wall


A PLASMOLYSED PLANT CELL

- If there is a potato cylinder that has not increased or decreased in mass, it means there was **no overall net movement of water** into or out of the potato cells
- This is because the solution that cylinder was in was the same concentration as the solution found in the cytoplasm of the potato cells, so there was **no concentration gradient**

Importance of Osmosis in Tissues

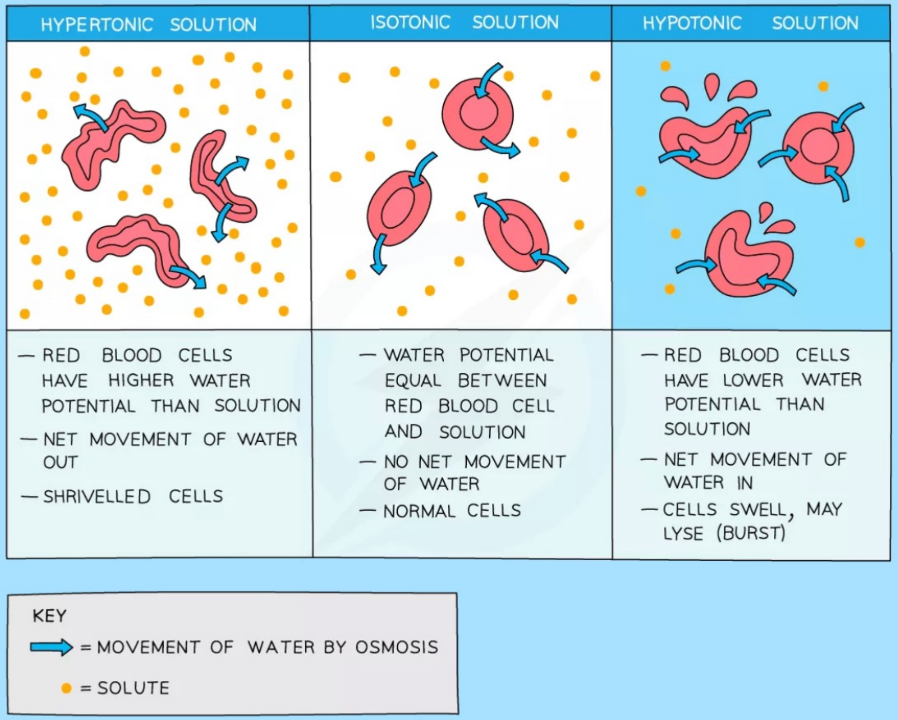
- When water moves into a plant cell, the vacuole gets bigger, **pushing the cell membrane against the cell wall**
- Water entering the cell by osmosis makes the cell **rigid and firm**
- This is important for plants as the effect of all the cells in a plant being firm is to **provide support and strength for the plant** – making the plant stand upright with its leaves held out to catch sunlight
- The pressure created by the **cell wall** stops too much water entering and **prevents the cell from bursting**
- If plants do not receive enough water the cells cannot remain rigid and firm (turgid) and the plant **wilts**

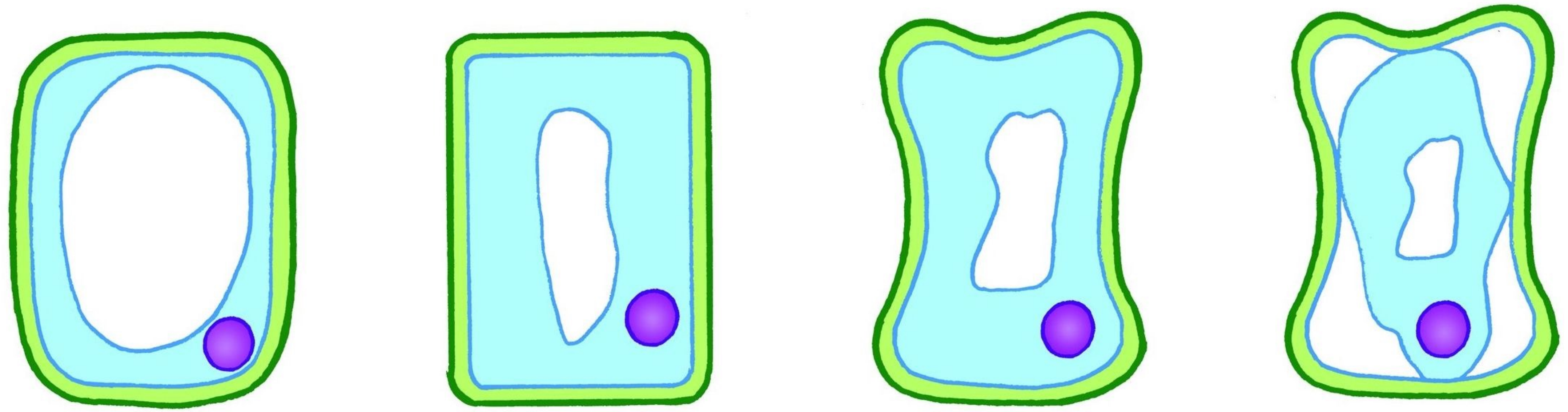
Osmosis in Plant Tissues

- Plant cells that are **turgid** are full of water and contain a **high turgor pressure** (the pressure of the cytoplasm pushing against the cell wall)
- This pressure **prevents any more water entering the cell** by osmosis, even if it is in a solution that has a higher water potential than inside the cytoplasm of the cells
- This **prevents** the plant cells from taking in too much water and **bursting**
- Plant roots are surrounded by soil water and the cytoplasm of root cells has a **lower water potential** than the soil water
- This means water will move across the cell membrane of root hair cells **into the root by osmosis**
- The water moves **across the root from cell to cell by osmosis** until it reaches the xylem Once they enter the **xylem** they are transported away from the root by the transpiration stream, helping to maintain a **concentration gradient** between the root cells and the xylem vessels

Osmosis in Animal Tissues

- **Animal cells** also lose and gain water as a result of osmosis
- As animal cells **do not have a supporting cell wall**, the results on the cell are more severe
- If an animal cell is placed into a **strong sugar solution** (with a lower water potential than the cell), it will lose water by osmosis and become **crenated** (shrivelled up)
- If an animal cell is placed into **distilled water** (with a higher water potential than the cell), it will gain water by osmosis and, as it has **no cell wall to create turgor pressure**, will continue to do so until the cell membrane is stretched too far and **it bursts**





Cell in dilute solution becomes turgid.

Cell in same concentration of solution.

Cell in concentrated solution becomes flaccid.

Plasmolysed cell - cytoplasm is pulled away from the wall.

Let's Practise 3.2

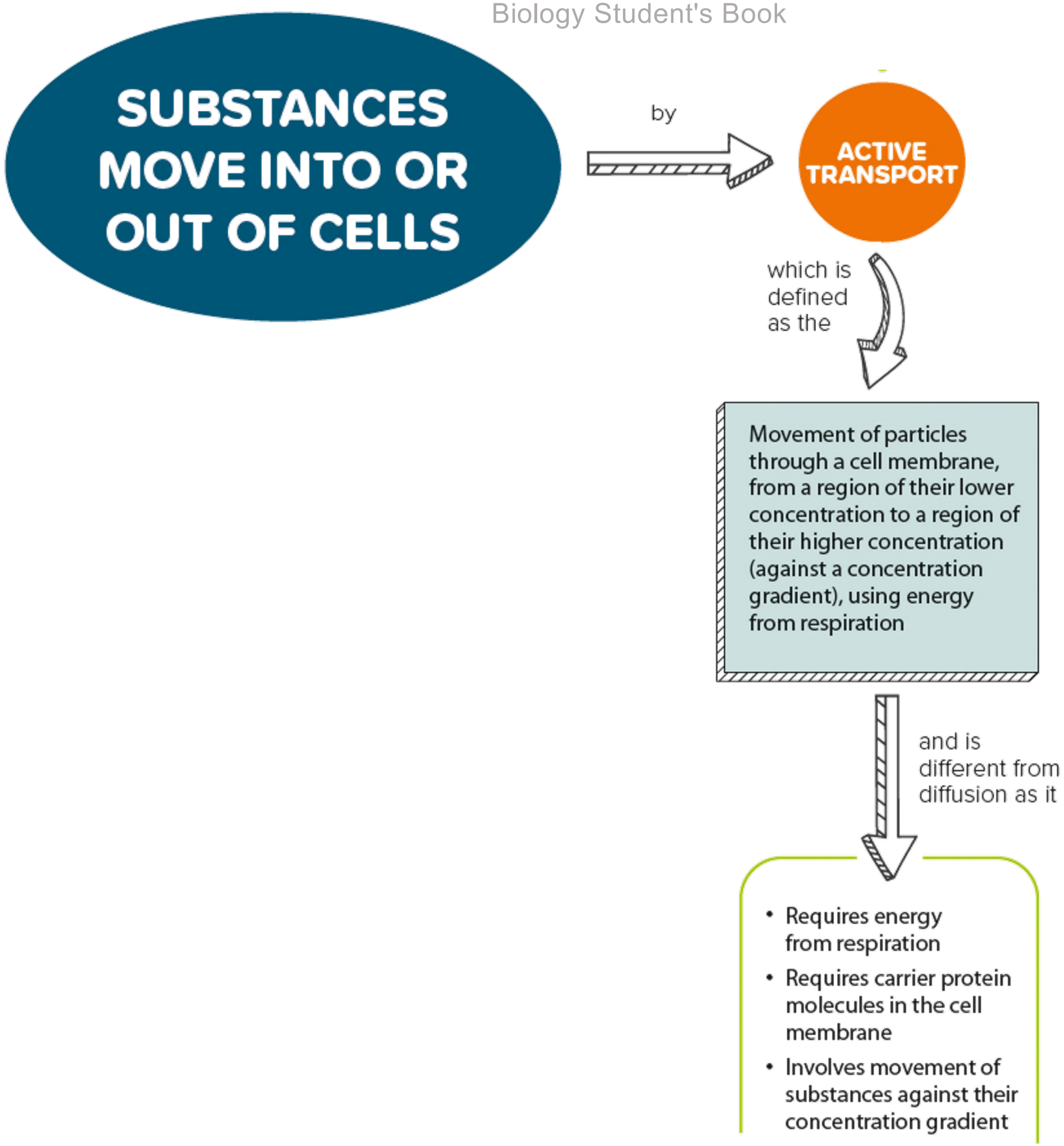
Humans have always needed to store food to see them through times when food is in short supply. Unfortunately, bacteria and fungi attack stored food and make it go bad. One way of preventing this is to store food in strong salt solutions (brine) or sugar solutions (syrups).

How does storing food in strong salt solutions or in syrups help to prevent the food from turning bad or decomposing?

3.3 Active Transport

In this section, you will learn the following:

- Describe active transport.
- S Explain the importance of active transport.
- S State that active transport requires protein carriers.



3.3 Active transport

Core

1 Describe active transport as the movement of particles through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient), using energy from respiration

Supplement

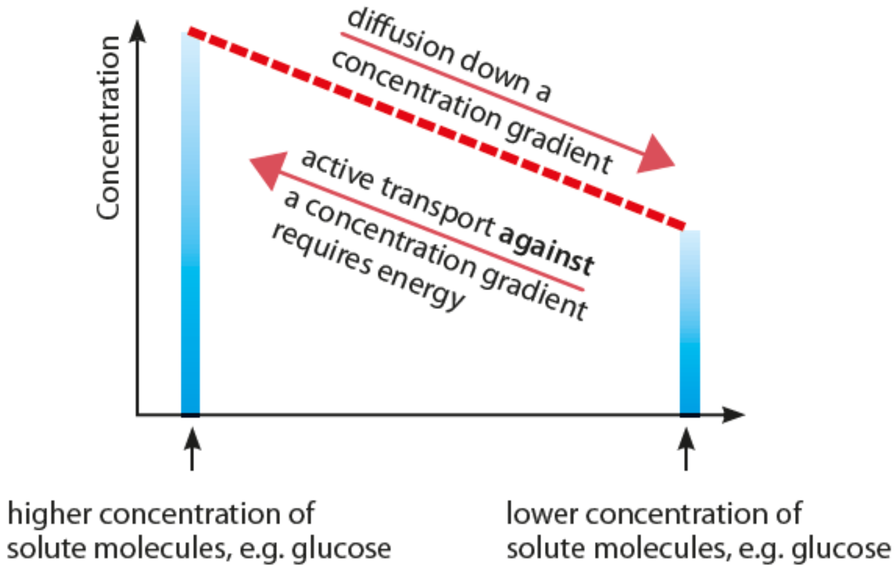
2 Explain the importance of active transport as a process for movement of molecules or ions across membranes, including ion uptake by root hairs

3 State that protein carriers move molecules or ions across a membrane during active transport

What are the differences between diffusion and active transport?

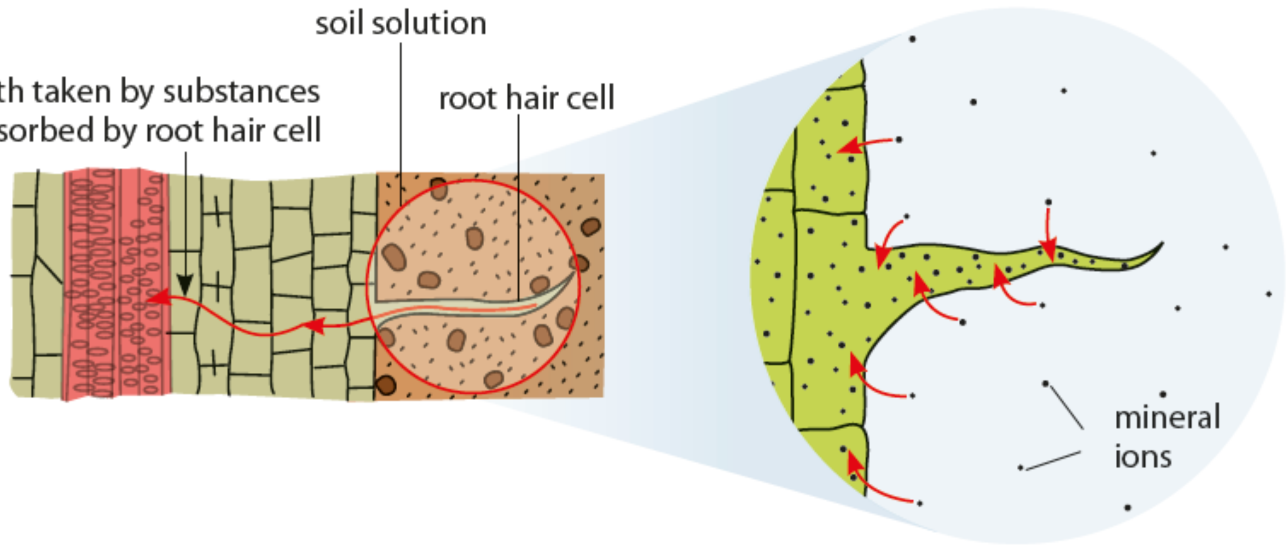
- **Active transport** is the movement of particles through a cell membrane, from a region of their lower concentration to a region of their higher concentration.
- Key differences between diffusion and active transport:

Diffusion	Active transport
Net movement of a substance, gaseous or solute	Movement of a dissolved solute
Movement of particles down a concentration gradient	Movement of particles against a concentration gradient
Energy from respiration not required	Energy from respiration required
Cell membrane not required	Cell membrane required



Where does active transport occur?

- Active transport occurs only in living cells because only living cells respire.
- In plants, root hair cells take in mineral ions from the soil solution through diffusion as well as active transport.

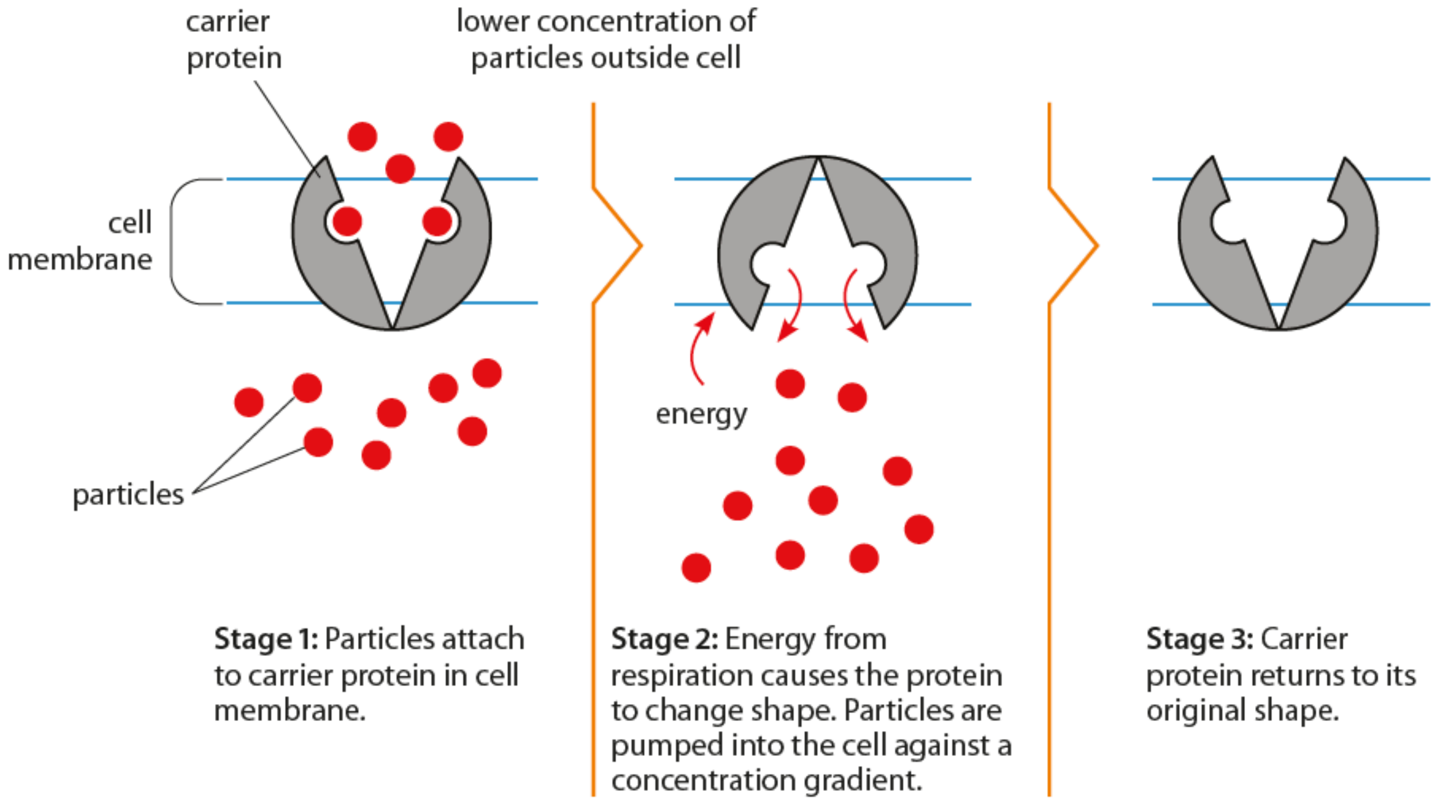


Mineral ions can enter the root hair cell, against the concentration gradient

LINK
How do root hair cells carry out the uptake of mineral ions? Find out more in Chapter 6.

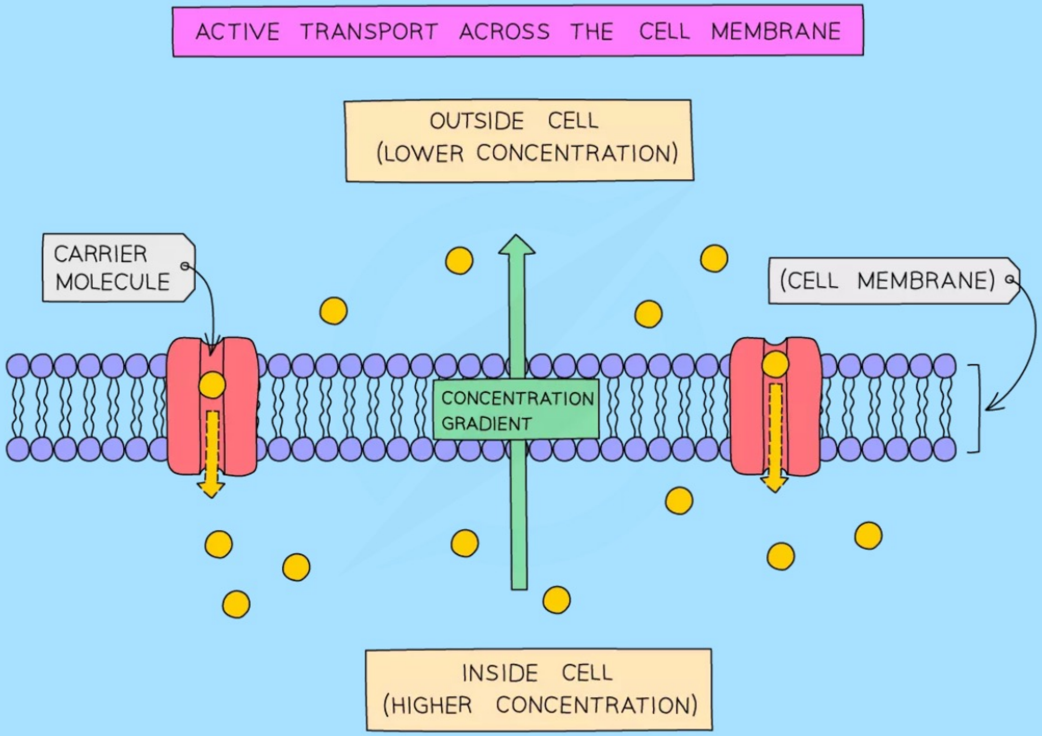
Where does active transport occur?

- Active transport requires carrier proteins in the cell membrane.



Examples of Active Transport

- Energy is needed because particles are being moved **against a concentration gradient**, in the opposite direction from which they would naturally move (by diffusion)
- Examples of active transport include:
  - **uptake of glucose** by epithelial cells in the villi of the small intestine and by kidney tubules in the nephron
  - **uptake of ions** from soil water by root hair cells in plants



Let's Practise 3.3

1. Why does active transport not take place in dead cells?
2. Name **three** characteristics that are different between diffusion and active transport.

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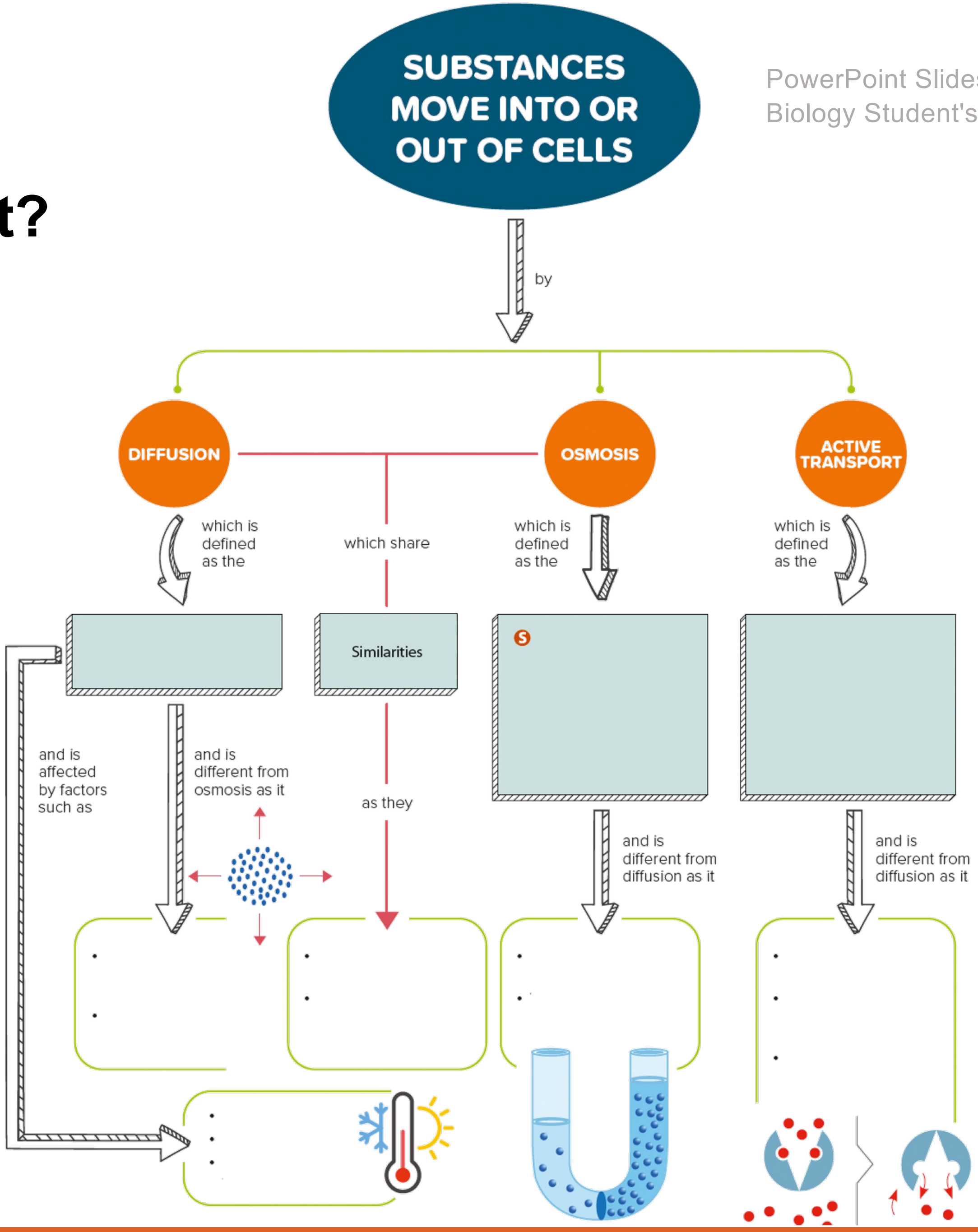
What have you learnt?



What have you learnt?

Can you draw your own mind map?